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SERIES OF TECHNICAL NOTES ON THE IMPACT OF NATURAL GAS DEFICIT AND THE INCREASE IN PRICES FOR END USERS: GENERAL PRESENTATION

Note 3. Macroeconomic costs of reducing the supply of natural gas

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Technical notes series on the impact of the natural gas shortage and price increases for end users: general overview

Due to a combination of past and present public policies, which have introduced transaction costs to the exploration and exploitation of conventional and unconventional domestic natural gas, indecisions on the use, size, location and remuneration of new import facilities, inflexibilities in contracting and charging by distance of the transport system, Colombia faces a deficit and possible shortage of natural gas in the short term.

Fedesarrollo presents four technical notes on the impacts of the natural gas deficit due to the reduction in domestic supply, the difficulties in expanding import capacities in the short term, and the increase in prices for end users due to import needs.
These notes aim to provide elements for decision-making within the State and to inform the public with public data and arguments guided by the general interest.

The titles of the technical notes are as follows:

Note 1. Increase in the price of residential natural gas in scenarios of greater imports.

Note 2. Fiscal cost of increased subsidies due to rising natural gas prices.

Note 3. Macroeconomic costs of reducing the supply of natural gas.

Note 4. Short and medium term public policy and regulation.

Note 5. Energy poverty due to reduced supply and increased price of natural gas.

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Note 3. Macroeconomic costs of reducing the supply of natural gas

Sara Ramírez, Carolina Silva, Nicolás Montoya, Sergio Cabrales and Juan Benavides (Fedesarrollo, Colombia)1 .

Objective and main message

Natural gas is essential for the efficient functioning of economic activity, especially in the industrial and commercial sectors, which depend on this input for their essential production processes. These two sectors account for 67% of the country's national demand for non-thermal natural gas and approximately 29% of the gross domestic product. High prices and restrictions in the supply of natural gas can generate significant negative impacts on the economy. Furthermore, this situation could lead users to return to coal, with its high environmental cost, or to resort to other, more expensive energy sources, such as liquefied petroleum gas (LPG) and diesel.

This technical note estimates (i) the short-term impacts of rising natural gas prices in a context of lower domestic production and higher imports on economic growth and the trade balance, and (ii) the medium-term effects of potential natural gas rationing in the country. The analysis focuses on industry, commerce, public utilities, mining, and the agricultural sector.

The results of the estimates are summarized below:

Between 2025 and 2031, without additions to domestic reserves, average GDP growth would be reduced by 0.1 percentage points as a result of the sustained increase in gas prices. This reduced dynamism would translate into a cumulative loss of $11.2 trillion over the next eight years. The higher gas prices would reduce average industrial growth by 0.6 percentage points annually, trade growth by 0.1 percentage points, and GDP growth by 0.1 percentage points.

From 2032 onward, without further additions to domestic reserves, the combination of increased tariffs and supply constraints would deepen the losses of the 2025-2031 period, reducing industrial growth by 1.4 percentage points and trade growth by 0.3 percentage points. These conditions would reduce the country's GDP by up to 0.4 percentage points in the medium term compared to the baseline scenario, from 2.9% to 2.5%.
of GDP.

ÿ On the external front, the current account deficit would widen, on average, by 0.2 percentage points of GDP annually from 2026 onwards, explained by an increase in gas imports to meet demand.

ÿ These projections imply a negative outlook for the next two governments, at least, due to the loss of growth, the competitiveness of industry, and the well-being of households (the latter is a matter already addressed in Note 2 of this series. It is a call to action with the recommendations that will be presented in Note 5).

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1 Disclaimer: The opinions expressed in this publication are those of the authors and should not be attributed to Fedesarrollo.

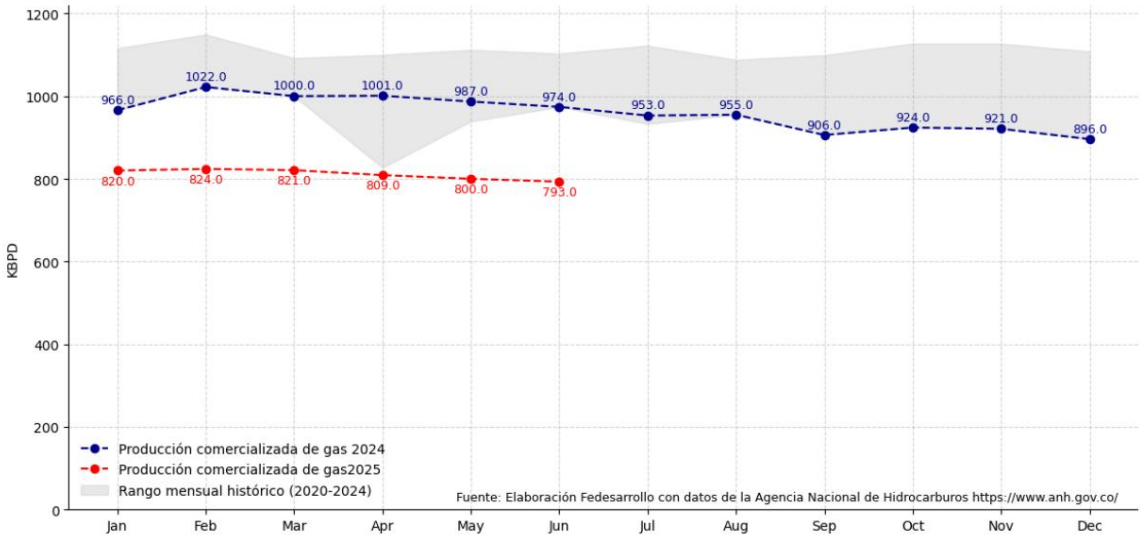
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Context and problem

Between 2012 and 2024, proven gas reserves decreased by 64%, falling from 5,727 gigapies cubic meters (GPC) in 2012 to 2,064 GPC in 2024. This has led to the self-sufficiency indicator, understood as the ratio between proven reserves and annual production (R/P), going from 13.4 to 5.9 years.

In the short term, domestic natural gas production has fallen. According to the National Hydrocarbons Agency (2025), 811.2 GBTU/day were produced in the first half of 2025, representing a 15.4% decrease compared to the 958.8 GBTU/day of 2024 and a 23.4% decrease compared to the 1,058.7 GBTU/day of 2023 (Figure 1). This is mainly due to the depletion of fields in La Guajira and the Llanos foothills.

Figure 1. Marketed production of natural gas (2020-2025)



Source: Prepared by Fedesarrollo with data from the National Hydrocarbons Agency (ANH).

The deterioration of reserves, the fall in national production and the increase in demand have modified the gas balance in Colombia, which went from being a self-sufficient market to depending increasingly on imports of liquefied natural gas (LNG) to meet domestic demand, initially in 2016 for thermal plants and, more recently, in 2024 for essential demand (residential, commercial, compression stations, vehicular natural gas and refineries).

According to DANE (2025), external purchases of natural gas represented 1.6% of total imports in 2024, compared to 0.1% in 2021. The growing dependence on imports faces structural limitations in the country's energy infrastructure, given that the El Cayao Port Society (SPEC) is currently the only operational regasification terminal in Colombia. During 2024, this plant registered an average utilization of 50.3% of its maximum capacity, so if the downward trend in production continues, the situation will worsen.

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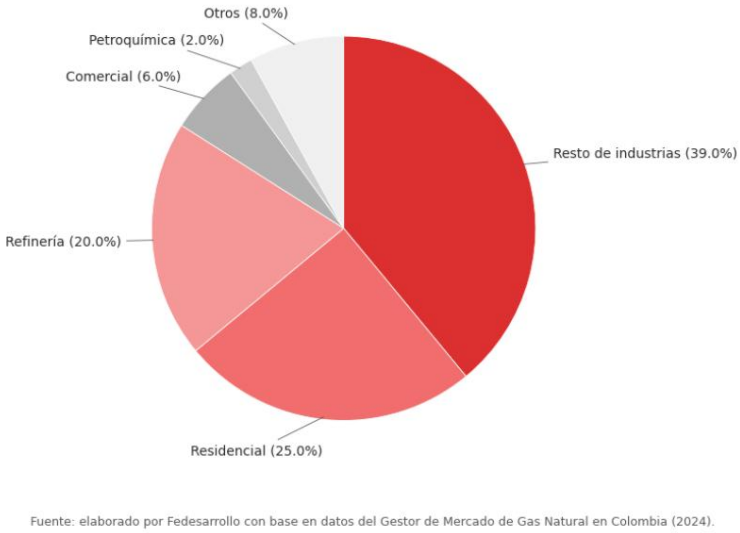
2 The main sources of natural gas imports during 2024 were the United States (US$ 2,790 million CIF, 87.9% of total natural gas imports), Trinidad and Tobago (US$ 305 million CIF, 9.6%), Tanzania (US$ 130 million CIF, 4.1%) and the United Kingdom (US$ 78 million CIF, 2.5%).

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Nationally, this capacity and future expansions could become completely saturated in the coming years. The vulnerability of the supply could be exacerbated by delays in the commissioning of the Buenaventura regasification plant and other import projects on the Caribbean Coast, such as Ballena and Coveñas.

Supply problems critically affect the industrial and commercial sectors, which account for 67% of the country's non-thermal natural gas demand (Graph 2).

Figure 2. Sectoral share of non-thermal natural gas demand (%) (2019-2024)



Source: Prepared by Fedesarrollo using data from the Natural Gas Market Manager in Colombia (2024).

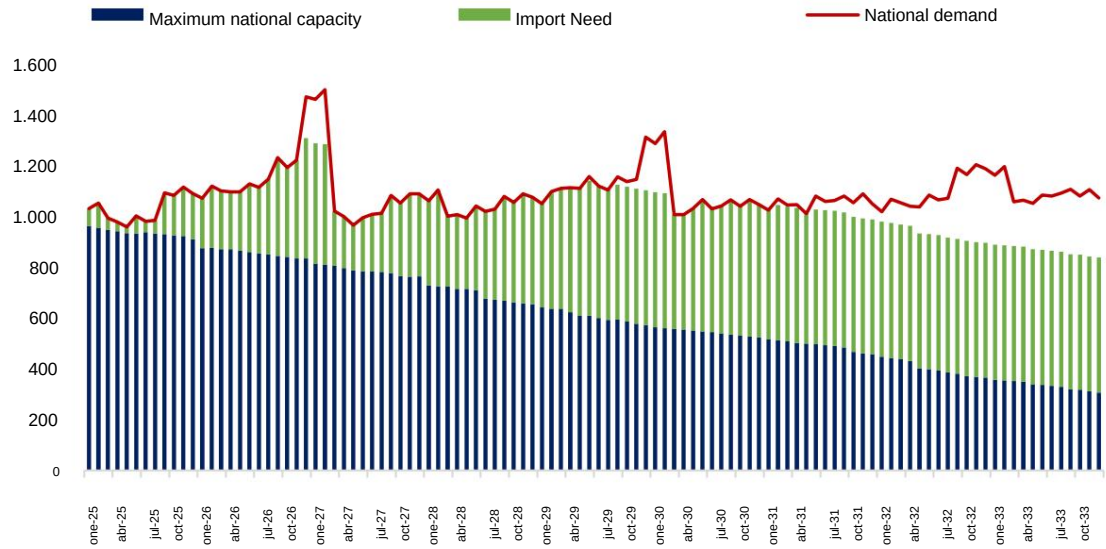
The industrial sector, particularly refining and petrochemicals, accounts for 38.8% of industrial value added and 22% of national non-thermal natural gas consumption. Tariff increases and potential rationing would generate significant losses in the industrial sector, which would then spread to other activities, weakening the productive fabric and jeopardizing economic growth. This, in turn, would promote the use of more polluting energy sources, such as coal, or more expensive ones, such as liquefied petroleum gas (LPG) or diesel.

Impacts and findings of Fedesarrollo

Natural gas supply projections (Figure 3) through 2033 show that the sum of expected domestic production and the planned increase in SPEC's regasification capacity, without new additions to domestic reserves or new import facilities, would generate an average natural gas deficit of 12.4% between December 2026 and February 2027, and 9.4% between September 2029 and February 2030, of a seasonal nature (May due to increased demand for thermal power generation during periods of low hydrology). Between 2025 and 2031, on average, 65.8% of potential natural gas demand would be met by domestic production, while the remaining 34.2% would depend on imports through regasification. However, from mid-2031 onwards, the system would face a structural deficit, even in scenarios without climate impacts: the total supply capacity would generate an average natural gas deficit of 3.5% of potential demand in 2031, 14.6% in 2032 and 21.0% in 2033.

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Figure 3. Average monthly gas supply with production potential (2024) and maximum regasification capacity (GBTUD)



Source: Prepared by Fedesarrollo based on calculations by ANH.

Imported gas would not only be insufficient to cover the local deficit—due to SPEC's limited capacity—but its price could also be up to three times higher than that of domestically produced gas in previous years. This would increase gas tariffs, by amounts calculated by Cabrales and Benavides (2025) in Technical Note 1 of this series.

The combination of rising gas prices and potential rationing would have adverse effects on national economic growth. The following section presents the impacts of these two factors—price increases and rationing—on economic activity, with a focus on the industrial and commercial sectors.

Impact of the increase in natural gas tariffs on industry, trade and total GDP

The increased need for gas imports will raise gas prices by an average of 32% compared to the national between 2025 and level observed in 2024 (Cabrales and Benavides).
20333 , 2025). From a sectoral perspective, this price increase would have a direct impact on industrial production —including sectors such as refining and petrochemicals that are difficult to decarbonize—, which accounts for about 65% of the demand for natural gas not intended for thermal generation.

Candelo, Oviedo, and Rivera (2023) analyze the effect of natural gas prices on industrial production in Colombia during the period 1997–2022. The results indicate that a

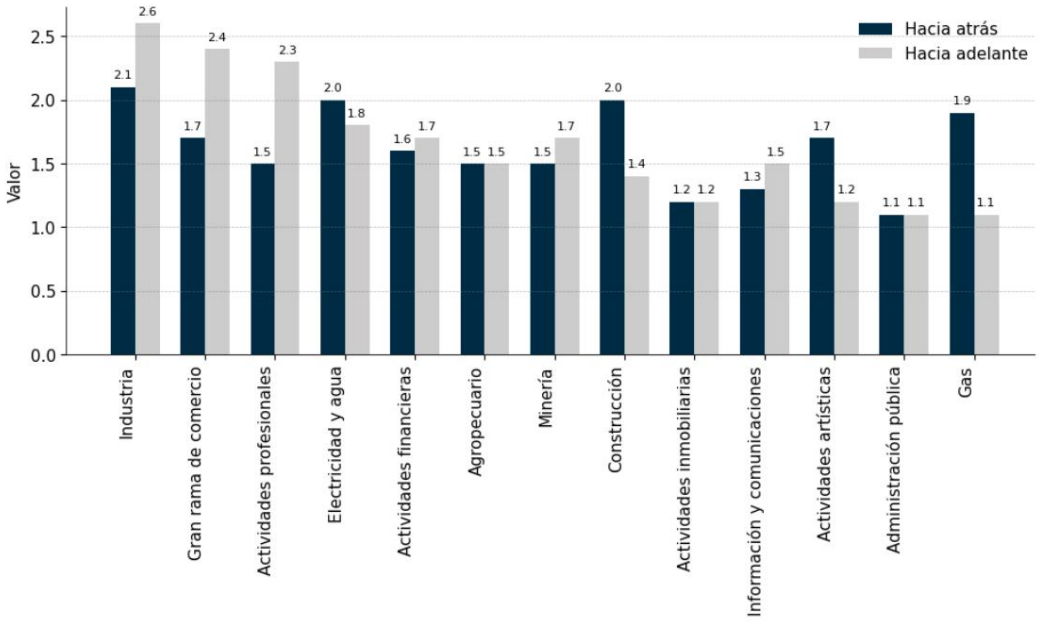
3 The increase in the gas tariff corresponds to the weighted average of the projected increases in natural gas tariffs in five of the country's main cities, as detailed in Cabrales and Benavides (2025). As mentioned there, in a scenario where 50% of the gas is imported, the tariff in Bogotá would increase by 44.6%, in Medellín by 44.9%, in Bucaramanga by 45.8%, in Cali by 30.3%, and in Barranquilla by 11.7%. The tariff increase in Cali is relatively smaller, since the city starts with a high base transportation tariff from Cusiana, which makes the increase due to gas imported from Cartagena less than in other cities.

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A 1% increase in the price of natural gas is associated with a 0.084% drop in industrial production. Using this elasticity, the average growth of the industrial sector would decrease by 0.6 percentage points between 2025 and 2033, falling from 1.9% in the central scenario to 1.3% in an environment of higher gas prices. As a result of this reduced dynamism, the industry would accumulate a loss of 6.9 trillion pesos over the next eight years, decreasing from $1,372.1 trillion to $1,365.2 trillion . Additionally, in 2033, the value added of industrial activities would be 5.4% below the level it would have reached without the increase in gas tariffs. This situation would hinder the recovery of the sector, which in the first quarter of 2025 registered a contraction of 0.4%, showing a slight improvement compared to the annual contractions of 2.7% and 2.1% observed in 2023 and 2024, respectively.

The contraction in industrial production would generate significant repercussions on economic activity due to its production linkages. Looking at the GDP multipliers of the input-output matrix, industry has the strongest linkages, both with sectors that supply inputs and intermediate goods (backward linkages) and with those that receive industrial products for subsequent use (forward linkages). For every additional peso of production in industry, a demand of 2.1 pesos is generated in the input-supplying sectors and 2.6 pesos in the sectors that receive industrial products (Figure 4).

Graph 4. GDP Multipliers



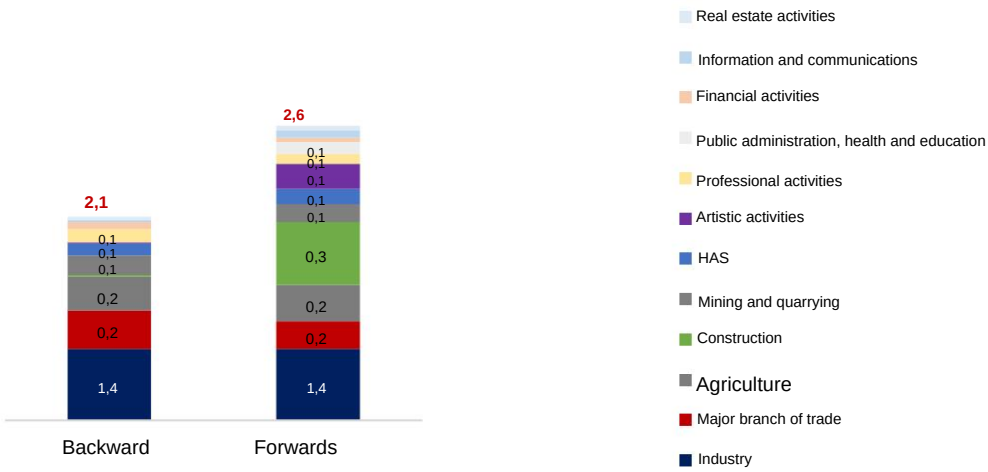
Source: Prepared by Fedesarrollo using data from DANE.

The industry has a multiplier of 1.4, highlighting its role as the main buyer and supplier of products within its own sector. Furthermore, the industry's strongest linkages with other sectors, both forward and backward, are with the major trade sector and agricultural activities, with a coefficient of 0.2 in both directions (Figure 5).

4 All values are expressed at constant 2025 prices.

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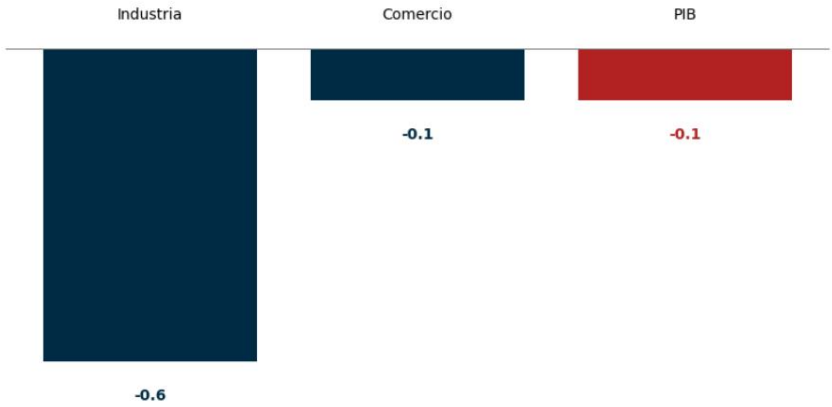
Figure 5. Production chains of the industry



Source: Prepared by Fedesarrollo using data from DANE.

Based on these results, and given the importance of trade within the industrial production structure, the effects of increased gas prices on the value added of the major trade sector, which represents approximately 18.9% of GDP, are estimated. It is projected that, between 2025 and 2033, the average growth of this activity would be 0.1 percentage points lower, falling from 3.4% in the central scenario to 3.3% in a context of higher gas prices and reduced industrial dynamism (Figure 6).

Chart 6. Impact of the increase in the gas tariff on growth 2025-2033 (annual percentage points)



Source: Prepared by Fedesarrollo using data from DANE.

The large trade sector would register a cumulative loss of $1.2 trillion compared to a scenario without gas tariff increases, with its value added falling from $2,130.8 trillion to $2,129.6 trillion over this period. This would imply that, in 2033, the value added of trade activities would be 0.6% below the level it would have reached in the absence of the gas tariff increase.

Taking into account the linkages with all sectors of the economy, it is estimated that, between 2025 and 2033, average GDP growth would be 0.1 percentage points lower as a result of the sustained increase in gas prices. This reduced dynamism would translate into

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a cumulative loss of $11.2 trillion over the next eight years, falling from an aggregate value of $12,440.9 trillion in the absence of the shock to $12,429.7 trillion. In particular, 72% of this loss in absolute terms would correspond to the industry and trade sectors, which would jointly register a cumulative loss of $8.1 trillion in the coming years (Table 1).

This would be followed by agricultural activities, the supply of electricity, gas and water, and mining, which would register losses of $0.9, $0.8 and $0.6 trillion, respectively.

Table 1. Cumulative GDP loss due to gas tariff increase between 2025 and 2033
(trillions of constant 2025 pesos)

	Base scenario	Increased scenario gas prices	Loss
Manufacturing industries	1.372,1	1.365,2	6,9
Trade	2.130,8	2.129,6	1,2
Agriculture	744,5	743,6	0,9
HAS	309,6	308,8	0,8
Mining	329,7	329,0	0,6
Rest	7.554,3	7.553,4	0,8
START	12.440,9	12,429.7	11,2

Source: Fedesarrollo elaboration with data from DANE.

From a demand perspective, the lower GDP growth between 2025 and 2033 would be primarily explained by a reduction in investment in machinery and equipment, whose average growth would decrease by 0.6 percentage points, from 4.8% to 4.2%, in line with the slower pace of industrial activity. Meanwhile, the increase in gas prices would generate additional inflationary pressures that could lead to a more contractionary monetary policy compared to a scenario without this shock, which would discourage private consumption, whose growth would decrease by 0.1 percentage points, from 2.5% to 2.4%, in line with the slower pace of trade.

On the external front, gas imports are projected to grow by 213% in 2026, driven by the effects of the El Niño phenomenon expected in the last quarter of the year. This increase would bring the imported volume to an average of 297 GBTU per day, and given that natural gas imports represent 1.6% of total external purchases of goods, it would translate into an additional increase of 1.4 percentage points in the annual growth of total goods imports. As a result, in 2026 the trade deficit in goods would widen by 0.2 percentage points, reaching 3.7% of GDP, as would the current account deficit, which would stand at 3.0% of GDP. In 2029, both the increase in regasification capacity at the Sociedad Portuaria El Cayao (SPEC) plant and the increase in gas demand towards the end of the year would lead to a 42% increase in external gas purchases, and would generate an additional increase of 0.3 percentage points in the annual growth of total goods imports, without a significant impact on the current account deficit5 .

5. In 2025, barring an El Niño event, natural gas imports are projected to fall by 58% compared to 2024, averaging 95 GBTU per day due to lower demand from thermoelectric power plants. In 2027, even without an El Niño event, imports are expected to decline by only 3% compared to 2026, due to the depletion of domestic natural gas reserves. This same factor would explain the 23% increase in 2028. In the years following 2029, growth is projected to be less than 5% annually, limited by maximum regasification capacity.

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Impact of gas rationing on industry, trade and GDP

In addition to rising natural gas prices, if domestic production and regasification capacity are not increased to allow for greater imports, the country could face rationing by late 2026 or late 2029 during the dry seasons, which require increased thermoelectric generation. Specifically, a structural natural gas deficit equivalent to 3.5% of potential demand is projected for 2031, widening to 14.6% in 2032 and reaching 21.0% in 2033.

This situation would impose high costs, not only for industry and commerce, but also for households and the economy as a whole. Additionally, this situation could lead users to return to the cheapest energy source, coal, albeit at a high environmental cost, or to resort to other, more expensive energy sources, such as liquefied petroleum gas (LPG) and
diesel.

The Mining and Energy Planning Unit (UPME 2023) estimates the costs of natural gas service interruptions, both scheduled and unscheduled, in terms of the well-being of residential, industrial, and commercial users. To do this, it uses a contingent valuation methodology based on surveys that measure willingness to pay to avoid service interruptions. This approach is complemented by the analysis of load curves by sector and time of day, which allows for determining the magnitude of rationing according to the time of day and the type of user affected. It also considers consumers' ability to replace natural gas with other energy sources or alternative processes; for example, some industries can temporarily substitute gas with coal, LPG, or diesel. Based on these elements, the specific rationing costs are estimated for each type of user.

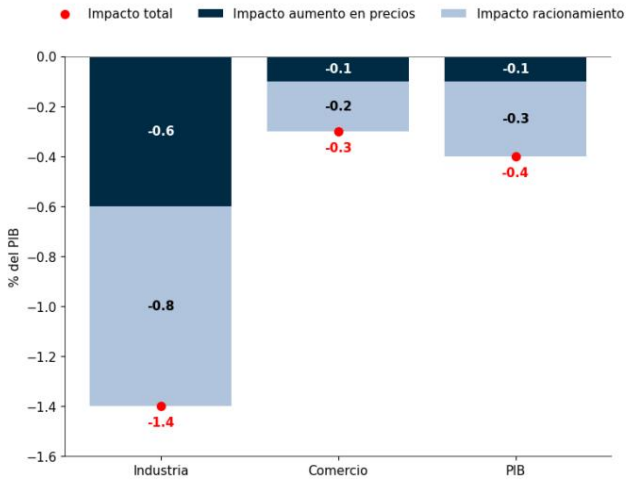
scenery.

Based on the projected natural gas deficit presented earlier, the percentage of demand that must be rationed is determined, through which the cost to be borne by the industrial and commercial sectors is estimated. Regarding the industrial sector, it is projected that in 2031 the losses generated would be marginal, while between 2032 and 2033 the average growth of the industrial sector would decrease by 1.4 percentage points, falling from 1.9% in the central scenario to 0.5%, under rationing and higher gas prices due to dependence on imports (Figure 7).

As a result, between 2032 and 2033, the industry would incur an additional loss of $11.4 trillion annually, on top of the costs already associated with the gas tariff increase ($6.9 trillion). In this scenario, its value added would decrease from $1,372.1 trillion in the baseline scenario to $1,353.8 trillion under rationing and higher gas tariff conditions (Table 2). Consequently, in 2033, the industry's value added would be 11.4% lower than the level it would have reached in the absence of both the tariff increase and the gas supply restrictions.

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Chart 7. Impact on growth 2032-2033 (annual percentage points)



Source: Prepared by Fedesarrollo using data from DANE.

Table 2. Total accumulated loss 2025 – 2032 (trillions of constant 2025 pesos)

	Scenery base	Loss increase prices (2025-2033)	Loss rationing (2032-2033)	Loss total	Scenery rationing and price increases
Industries	1.372,1	6,9	11,4	18,3	1.353,8
Trade	2.130,8	1,2	5,4	6,6	2.124,2
Agriculture	744,5	0,9	0,0	0,9	743,6
HAS	309,6	0,8	0,0	0,8	308,8
Mining	329,7	0,6	0,0	0,6	329,0
Rest	7.554,3	0,8	0,1	0,8	7.553,5

Source: Prepared by Fedesarrollo using data from DANE.

The large trade sector would experience an average reduction of 0.3 percentage points between 2032 and 2033, falling from 3.3% in the baseline scenario to 3.0% in a rationing scenario, which already incorporates the effects of increased natural gas prices and lower industrial production. Should this scenario materialize, trade would face an additional loss of $5.4 trillion during those years, on top of the effects already observed from the increase in natural gas tariffs ($1.2 trillion). Thus, the value added of the large trade sector would decrease from $2,130.8 trillion in the baseline scenario to $2,124.2 trillion under rationing conditions (Table 2). Consequently, in 2033, the value added of the trade sector would be, on average, 2.6% below the level it would have reached in the absence of both the tariff increase and the restrictions on natural gas supply.

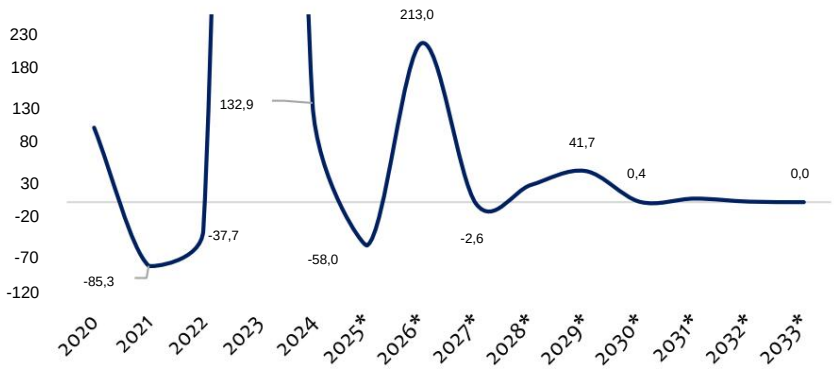
Due to the convergence of rising gas prices and increased rationing costs, average economic growth between 2032 and 2033 would be 0.4 percentage points lower than the baseline scenario, falling from 2.9% to 2.5%. Consequently, GDP would decrease by an additional $16.7 trillion during this period, on top of the loss associated with the gas tariff increase ($11.3 trillion), implying a total loss of $28 trillion, dropping from $12,440.9 trillion to $12,412.9 trillion, and placing it 3.6% below the level it would have reached.

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In the absence of these restrictions. From a demand perspective, the GDP slowdown between 2032 and 2033 would be mainly explained by a slower expansion of private consumption, which would decrease from 2.5% to 2.2%, and by a moderation in investment in machinery and equipment, whose growth would be reduced by 1.3 percentage points, from 4.8% to 3.5%.

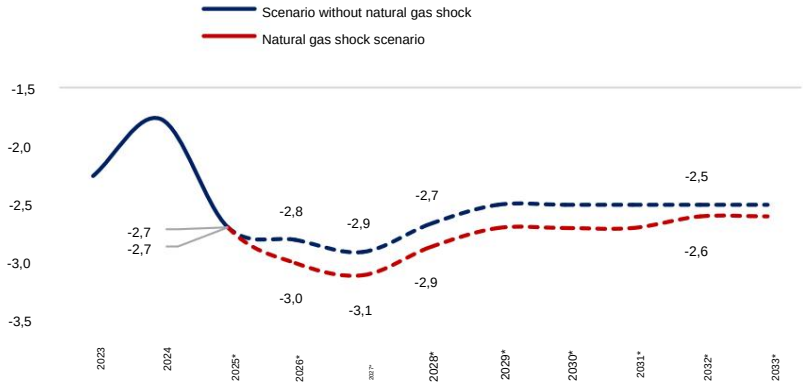
On the external front, the growing dependence on gas imports and the effects of rationing would generate differentiated impacts in the short and medium term. In the short term, gas imports are projected to increase by 213% in 2026, widening the current account deficit by 0.2 percentage points of GDP compared to a scenario in which domestic production is sufficient to meet internal demand. In the medium term, once the country reaches its maximum regasification capacity, gas imports would cease to grow, and the impact on the external accounts would stem from the slowdown in domestic demand under a rationing scenario in 2032 and 2033. Under this scenario, the growth of goods imports is estimated to decrease by 0.7 percentage points, allowing for a 0.1 percentage point reduction in GDP in both the goods trade deficit and the current account deficit (Figures 8 and 9).

Chart 8. Natural gas imports (annual change, %)



Source: Fedesarrollo analysis using data from the Central Bank of Colombia. *Projections Fedesarrollo.

Chart 9. Current account balance (% of GDP)



Source: Fedesarrollo analysis using data from the Central Bank of Colombia. *Fedesarrollo projections.

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